Data Analysis
AMS315

Chapter 5

Some Important Discrete Probability Distributions

Learning Objectives

In this chapter, you learn:

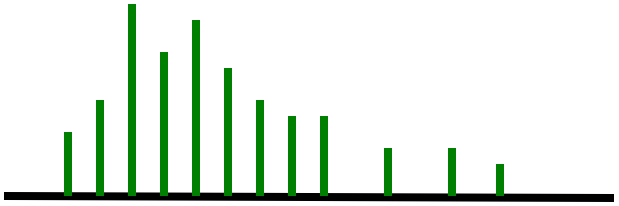
- The properties of a probability distribution
- To calculate the expected value and variance of a probability distribution
- To calculate the covariance and its use in finance
- To calculate probabilities from binomial, hypergeometric, and Poisson distributions
- How to use the binomial, hypergeometric, and Poisson distributions to solve business problems

Introduction to Probability Distributions

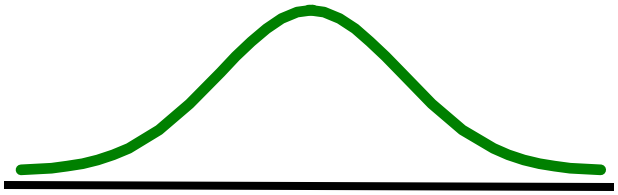
- **Random Variable**
  - Represents a possible numerical value from an uncertain event

Random Variables

Ch. 5 **Discrete Random Variable**



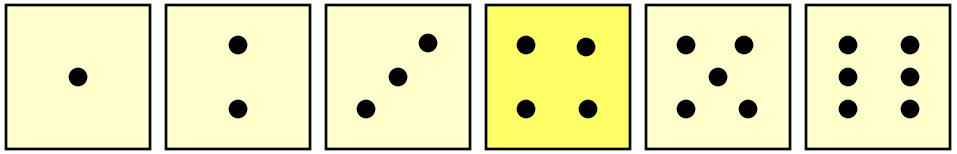
Continuous Random Variable Ch. 6



Discrete Random Variables

- Can only assume a countable number of values

Examples:



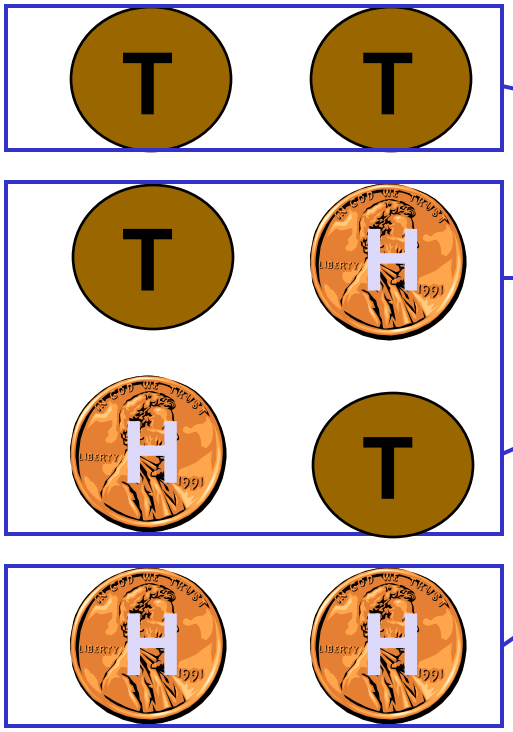
- **Roll a die twice**
  Let X be the number of times 4 comes up
  (then X could be 0, 1, or 2 times)

- **Toss a coin 5 times.**
  Let X be the number of heads
  (then X = 0, 1, 2, 3, 4, or 5)

Discrete Probability Distribution

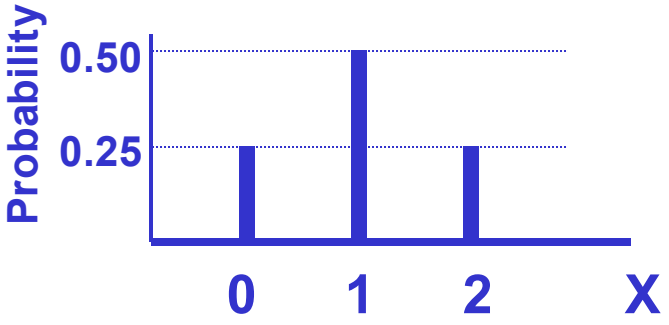
Experiment: Toss 2 Coins. Let X = # heads.

4 possible outcomes



Probability Distribution

X Value	Probability
0	1/4 = 0.25
1	2/4 = 0.50
2	1/4 = 0.25



Discrete Random Variable Summary Measures

- Expected Value (or mean) of a discrete distribution (Weighted Average)

$$\mu = E(X) = \sum_{i=1}^{N} X_i P(X_i)$$

- Example: Toss 2 coins,
  X = # of heads,
  compute expected value of X:

X	P(X)
0	0.25
1	0.50
2	0.25

E(X) =

Discrete Random Variable Summary Measures

(continued)

- Variance of a discrete random variable

$$\sigma^2 = \sum_{i=1}^{N} [X_i - E(X)]^2 P(X_i)$$

- Standard Deviation of a discrete random variable

$$\sigma = \sqrt{\sigma^2} = \sqrt{\sum_{i=1}^{N} [X_i - E(X)]^2 P(X_i)}$$

where:

$E(X)$ = Expected value of the discrete random variable X

X_i = the ith outcome of X

$P(X_i)$ = Probability of the ith occurrence of X

Discrete Random Variable Summary Measures

(continued)

- Example: Toss 2 coins, X = # heads, compute standard deviation (recall E(X) = 1)

$$\sigma = \sqrt{\sum [X_i - E(X)]^2 P(X_i)}$$

$$\sigma = \sqrt{(0-1)^2(0.25) + (1-1)^2(0.50) + (2-1)^2(0.25)} = \sqrt{0.50} = 0.707$$

Possible number of heads = 0, 1, or 2

The Covariance

- The covariance measures the strength of the linear relationship between two variables
- The covariance:

$$\sigma_{XY} = \sum_{i=1}^{N} [X_i - E(X)][(Y_i - E(Y)]\, P(X_i Y_i)$$

where: X = discrete variable X
X_i = the ith outcome of X
Y = discrete variable Y
Y_i = the ith outcome of Y
$P(X_i Y_i)$ = probability of occurrence of the ith outcome of X and the ith outcome of Y

Computing the Mean for Investment Returns

Return per $1,000 for two types of investments

$P(X_i Y_i)$	Economic condition	Investment	
		Passive Fund X	Aggressive Fund Y
0.2	Recession	- $ 25	- $200
0.5	Stable Economy	+ 50	+ 60
0.3	Expanding Economy	+ 100	+ 350

$E(X) = \mu_X = (-25)(0.2) + (50)(0.5) + (100)(0.3) = 50$

$E(Y) = \mu_Y =$

Computing the Standard Deviation for Investment Returns

$P(X_iY_i)$	Economic condition	Investment	
		Passive Fund X	Aggressive Fund Y
0.2	Recession	- $ 25	- $200
0.5	Stable Economy	+ 50	+ 60
0.3	Expanding Economy	+ 100	+ 350

$$\sigma_X = \sqrt{(-25-50)^2(0.2)+(50-50)^2(0.5)+(100-50)^2(0.3)}$$
$$= 43.30$$

$$\sigma_Y = \sqrt{(-200-95)^2(0.2)+(60-95)^2(0.5)+(350-95)^2(0.3)}$$
$$= 193.71$$

Computing the Covariance for Investment Returns

$P(X_iY_i)$	Economic condition	Investment	
		Passive Fund X	Aggressive Fund Y
0.2	Recession	- $ 25	- $200
0.5	Stable Economy	+ 50	+ 60
0.3	Expanding Economy	+ 100	+ 350

$$\sigma_{XY} = (-25-50)(-200-95)(0.2)+(50-50)(60-95)(0.5)$$
$$+(100-50)(350-95)(0.3)$$
$$= 8250$$

Interpreting the Results for Investment Returns

- The aggressive fund has a higher expected return, but much more risk

$$\mu_Y = 95 > \mu_X = 50$$

but

$$\sigma_Y = 193.71 > \sigma_X = 43.30$$

- The Covariance of 8250 indicates that the two investments are positively related and will vary in the same direction

The Sum of Two Random Variables

- Expected Value of the sum of two random variables:

$$E(X+Y)=E(X)+E(Y)$$

- Variance of the sum of two random variables:

$$Var(X+Y)=\sigma_{X+Y}^2=\sigma_X^2+\sigma_Y^2+2\sigma_{XY}$$

- Standard deviation of the sum of two random variables:

$$\sigma_{X+Y}=\sqrt{\sigma_{X+Y}^2}$$

Portfolio Expected Return and Portfolio Risk

- Portfolio expected return (weighted average return):

$$E(P) = wE(X) + (1 - w)E(Y)$$

- Portfolio risk (weighted variability)

$$\sigma_P = \sqrt{w^2\sigma_X^2 + (1 - w)^2\sigma_Y^2 + 2w(1 - w)\sigma_{XY}}$$

Where w = portion of portfolio value in asset X

(1 - w) = portion of portfolio value in asset Y

Portfolio Example

Investment X:	$\mu_X = 50$	$\sigma_X = 43.30$
Investment Y:	$\mu_Y = 95$	$\sigma_Y = 193.21$
	$\sigma_{XY} = 8250$	

Suppose 40% of the portfolio is in Investment X and 60% is in Investment Y, calculate:

(a) Portfolio expected return, and

(b) Portfolio risk

Note: The portfolio return and portfolio variability are between the values for investments X and Y considered individually

Probability Distributions

Probability Distributions

Ch. 5

Discrete Probability Distributions

- Binomial
- Poisson
- Hypergeometric

Continuous Probability Distributions

Ch. 6

- Normal
- Uniform
- Exponential

The Binomial Distribution

Probability Distributions

- **Discrete** Probability Distributions
  - Binomial
  - Poisson
  - Hypergeometric

Binomial Probability Distribution

- The binomial probability distribution is one of the most useful mathematical models.
- You use the binomial distribution when the discrete random variable of interest is the number of successes in a sample of n observations.
- The binomial distribution has four essential properties:

Binomial Probability Distribution

1. The sample consists of a fixed number of observations, *n*
   - e.g., 15 tosses of a coin; ten light bulbs taken from a warehouse
2. Each observation is classified into two mutually exclusive and collectively exhaustive categories
   - e.g., head or tail in each toss of a coin; defective or not defective light bulb
   - Generally called “success” and “failure”
   - Probability of success is *p* probability of failure is *1 – p*
3. Each observation has constant probability
   - e.g., Probability of getting a tail is the same each time we toss the coin

Binomial Probability Distribution

(continued)

4. Observations are independent
   - The outcome of one observation does not affect the outcome of the other

- Two sampling methods
  - Infinite population without replacement
  - Finite population with replacement

Possible Binomial Distribution Observations

- A manufacturing plant labels items as either defective or acceptable
- A firm bidding for contracts will either win or lose the contract
- A marketing research firm receives survey responses of “yes I will buy” or “no I will not”
- New job applicants either accept the offer or reject it

Rule of Combinations

- The number of combinations of selecting X objects out of n objects is

$$\boxed{{}_nC_x = \frac{n!}{X!(n-X)!}}$$

where:

$n! = (n)(n-1)(n-2)\ldots(2)(1)$

$X! = (X)(X-1)(X-2)\ldots(2)(1)$

$0! = 1$ (by definition)

Binomial Distribution Formula

$$P(X) = \frac{n!}{X!(n-X)!} p^{X} (1-p)^{n-X}$$

P(X) = probability of **X** successes in **n** trials, with probability of success **p** on each trial

X = number of 'successes' in sample, (X = 0, 1, 2, ..., n)

n = sample size (number of trials or observations)

p = probability of "success"

Example: Flip a coin four times, let x = # heads:

n = 4

p = 0.5

1 - p = (1 - 0.5) = 0.5

X = 0, 1, 2, 3, 4

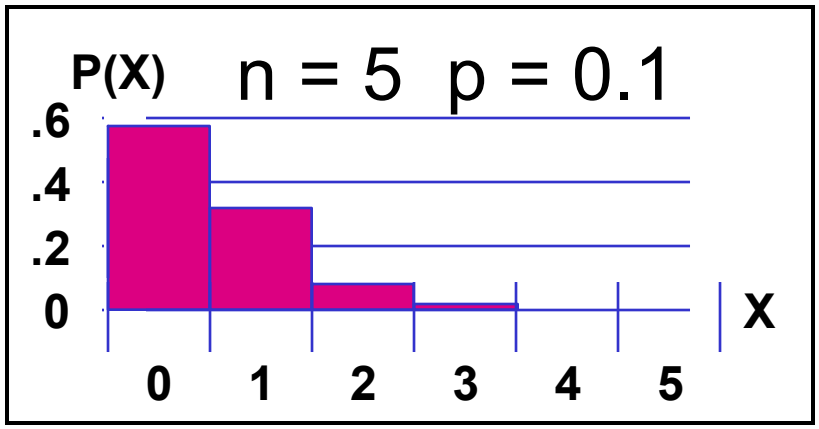
Example: Calculating a Binomial Probability

What is the probability of one success in five observations if the probability of success is 0.1?

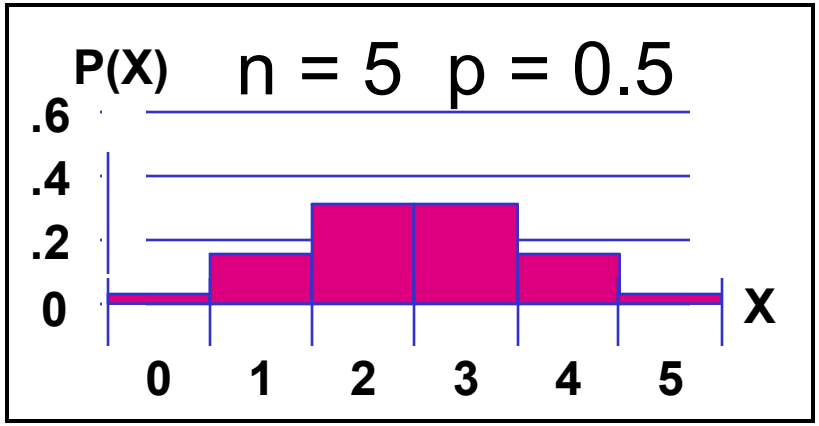
$X = 1$, $n = 5$, and $p = 0.1$

Binomial Distribution

- The shape of the binomial distribution depends on the values of p and n
  - Here, n = 5 and p = 0.1



  - Here, n = 5 and p = 0.5



Binomial Distribution Characteristics

- Mean

$$\mu = E(x) = np$$

- Variance and Standard Deviation

$$\sigma^2 = np(1-p)$$

$$\sigma = \sqrt{np(1-p)}$$

Where n = sample size

p = probability of success

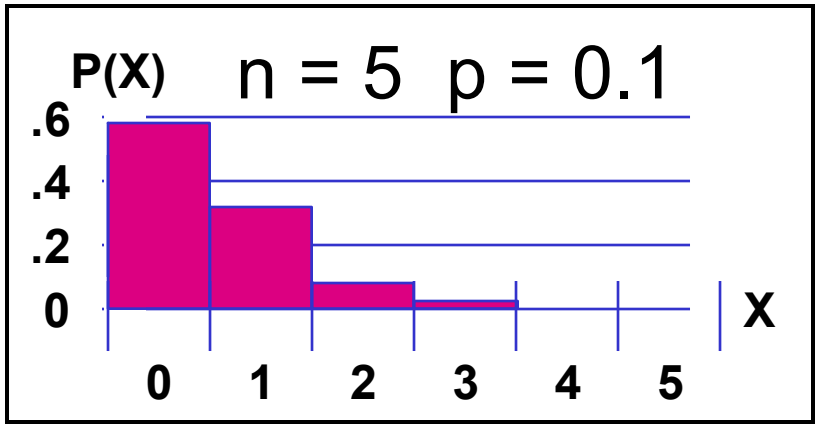
(1 – p) = probability of failure

Binomial Characteristics

Examples

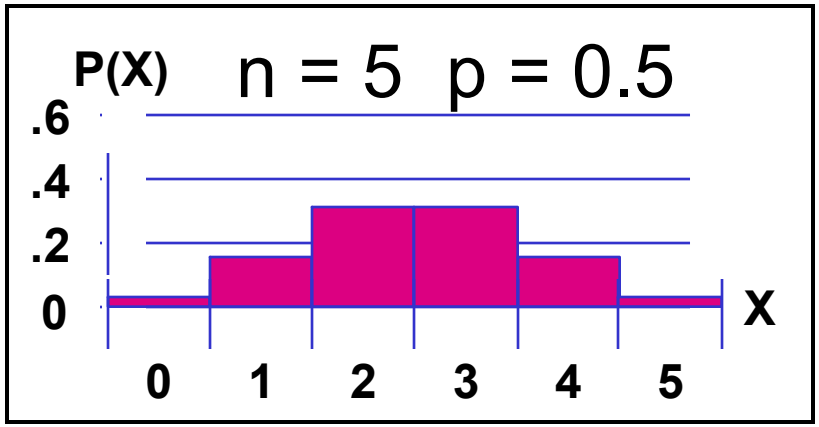
$$\mu = np = (5)(0.1) = 0.5$$

$$\sigma = \sqrt{np(1-p)} = \sqrt{(5)(0.1)(1-0.1)}$$
$$= 0.6708$$



$$\mu = np = (5)(0.5) = 2.5$$

$$\sigma = \sqrt{np(1-p)} = \sqrt{(5)(0.5)(1-0.5)}$$
$$= 1.118$$



Using Binomial Tables

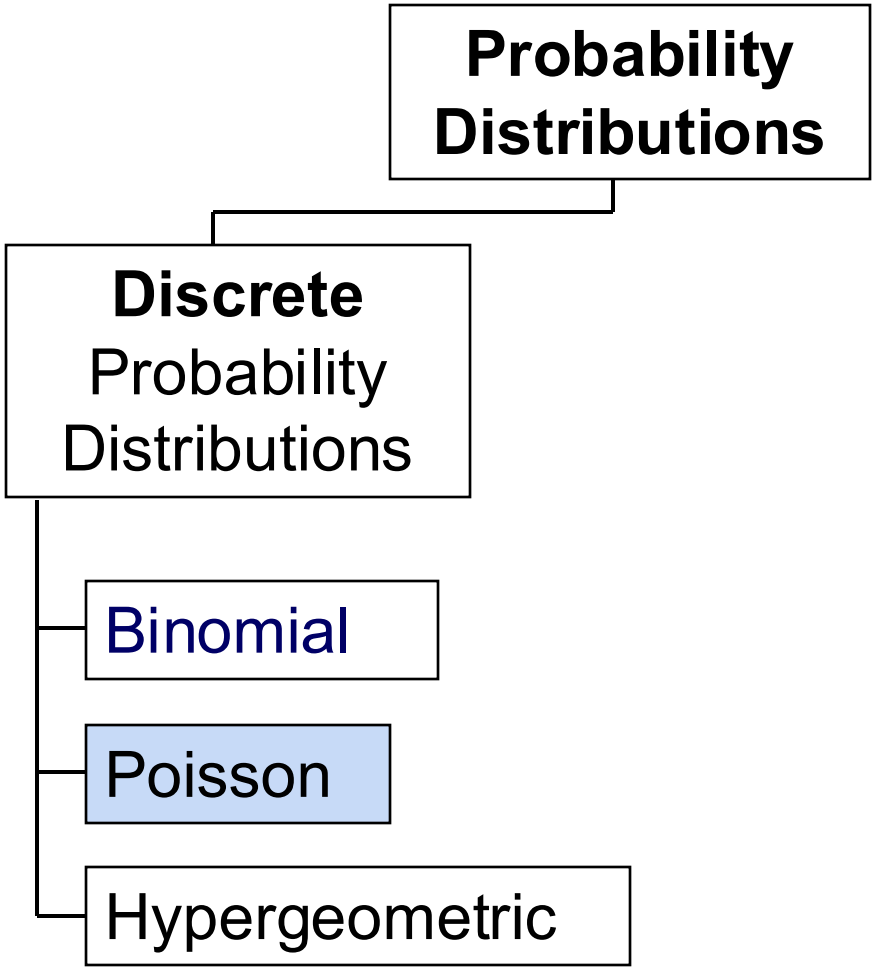
n = 10									
x	…	p=.20	p=.25	p=.30	**p=.35**	p=.40	p=.45	p=.50	
0	…	0.1074	0.0563	0.0282	0.0135	0.0060	0.0025	0.0010	10
1	…	0.2684	0.1877	0.1211	0.0725	0.0403	0.0207	0.0098	9
2	…	0.3020	0.2816	0.2335	0.1757	0.1209	0.0763	0.0439	8
3	…	0.2013	0.2503	0.2668	**0.2522**	0.2150	0.1665	0.1172	7
4	…	0.0881	0.1460	0.2001	0.2377	0.2508	0.2384	0.2051	6
5	…	0.0264	0.0584	0.1029	0.1536	0.2007	0.2340	0.2461	5
6	…	0.0055	0.0162	0.0368	0.0689	0.1115	0.1596	0.2051	4
7	…	0.0008	0.0031	0.0090	0.0212	0.0425	0.0746	0.1172	3
8	…	0.0001	**0.0004**	0.0014	0.0043	0.0106	0.0229	0.0439	**2**
9	…	0.0000	0.0000	0.0001	0.0005	0.0016	0.0042	0.0098	1
10	…	0.0000	0.0000	0.0000	0.0000	0.0001	0.0003	0.0010	0
	…	p=.80	**p=.75**	p=.70	p=.65	p=.60	p=.55	p=.50	**x**

Examples:

n = 10, p = 0.35, x = 3: $P(x = 3|n = 10, p = 0.35) = 0.2522$

n = 10, p = 0.75, x = 2: $P(x = 2|n = 10, p = 0.75) = 0.0004$

The Poisson Distribution



The Poisson Distribution

- Apply the Poisson Distribution when:
  - You wish to count the number of successes in an area of opportunity
  - The probability that an event occurs in one area of opportunity is the same for all areas of opportunity
  - The number of events that occur in one area of opportunity is independent of the number of events that occur in the other areas of opportunity
  - The probability that two or more events occur in an area of opportunity approaches zero as the area of opportunity becomes smaller
  - The average successes per unit is λ (lambda)

Poisson Distribution Formula

$$P(X) = \frac{e^{-\lambda}\lambda^{x}}{X!}$$

where:

X = number of success events in an area of opportunity

λ = average number of successes (mean)

e = base of the natural logarithm system (2.71828...)

Poisson Distribution Characteristics

- Mean

$$\mu = \lambda$$

- Variance and Standard Deviation

$$\sigma^2 = \lambda$$

$$\sigma = \sqrt{\lambda}$$

where λ = expected number of events

Using Poisson Tables

X	λ								
	0.10	0.20	0.30	0.40	**0.50**	0.60	0.70	0.80	0.90
0	0.9048	0.8187	0.7408	0.6703	0.6065	0.5488	0.4966	0.4493	0.4066
1	0.0905	0.1637	0.2222	0.2681	0.3033	0.3293	0.3476	0.3595	0.3659
2	0.0045	0.0164	0.0333	0.0536	**0.0758**	0.0988	0.1217	0.1438	0.1647
3	0.0002	0.0011	0.0033	0.0072	0.0126	0.0198	0.0284	0.0383	0.0494
4	0.0000	0.0001	0.0003	0.0007	0.0016	0.0030	0.0050	0.0077	0.0111
5	0.0000	0.0000	0.0000	0.0001	0.0002	0.0004	0.0007	0.0012	0.0020
6	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001	0.0002	0.0003
7	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Example: Find $P(X = 2)$ if $\lambda = 0.50$

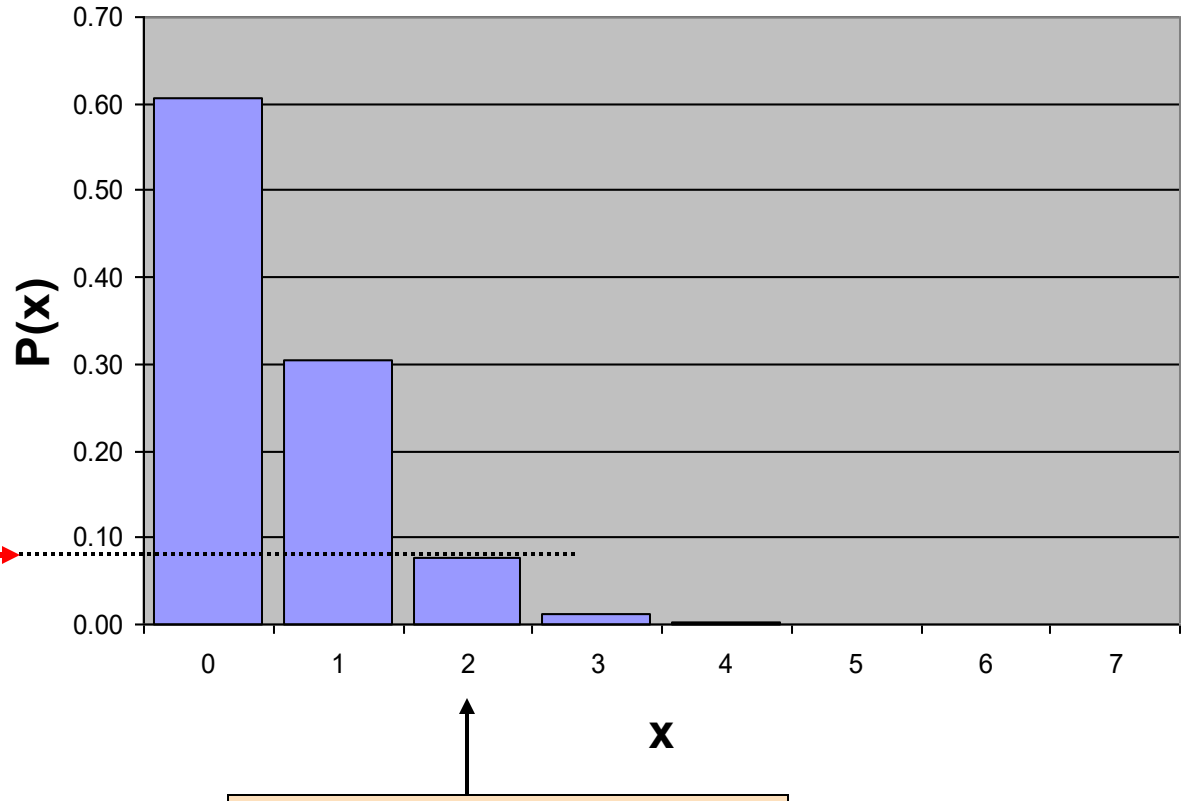
$$P(X=2)=\frac{e^{-\lambda}\lambda^{X}}{X!}=\frac{e^{-0.50}(0.50)^{2}}{2!}=0.0758$$

Graph of Poisson Probabilities

Graphically:

$\lambda = 0.50$

X	λ = 0.50
0	0.6065
1	0.3033
2	0.0758
3	0.0126
4	0.0016
5	0.0002
6	0.0000
7	0.0000

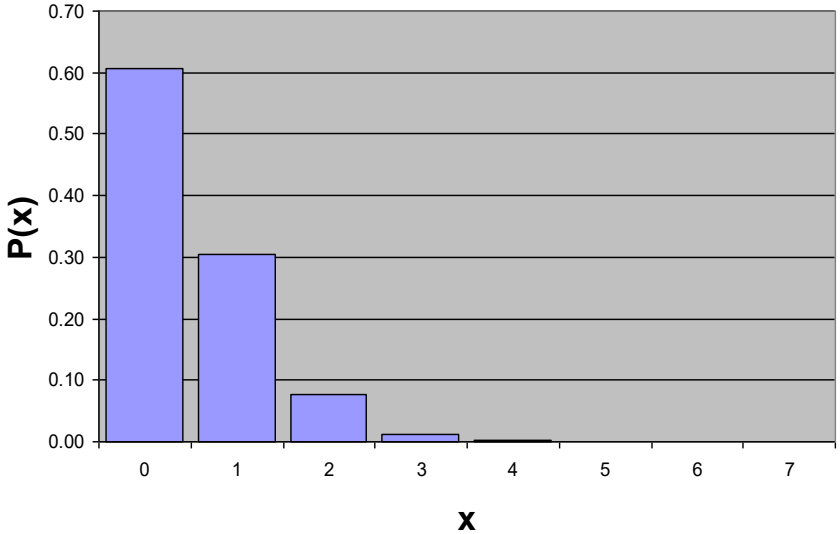


P(X = 2) = 0.0758

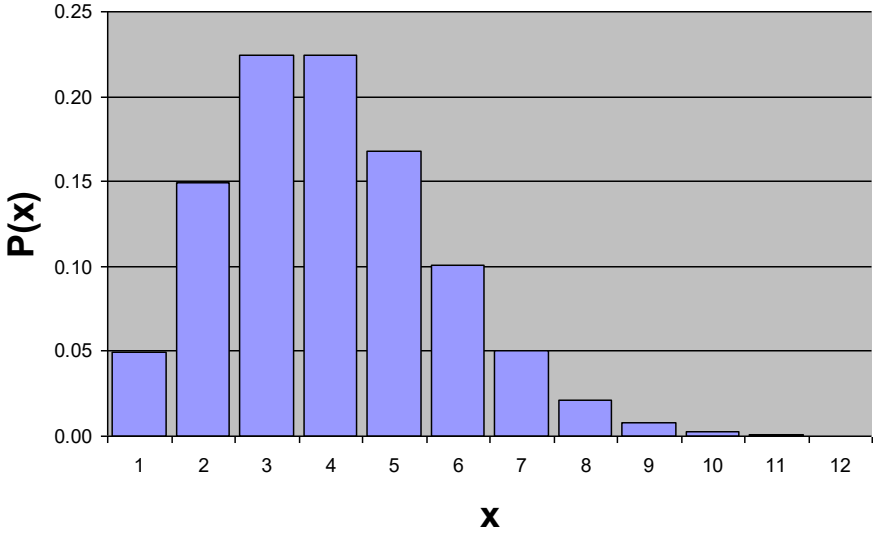
Poisson Distribution Shape

- The shape of the Poisson Distribution depends on the parameter λ :

$\lambda = 0.50$



$\lambda = 3.00$

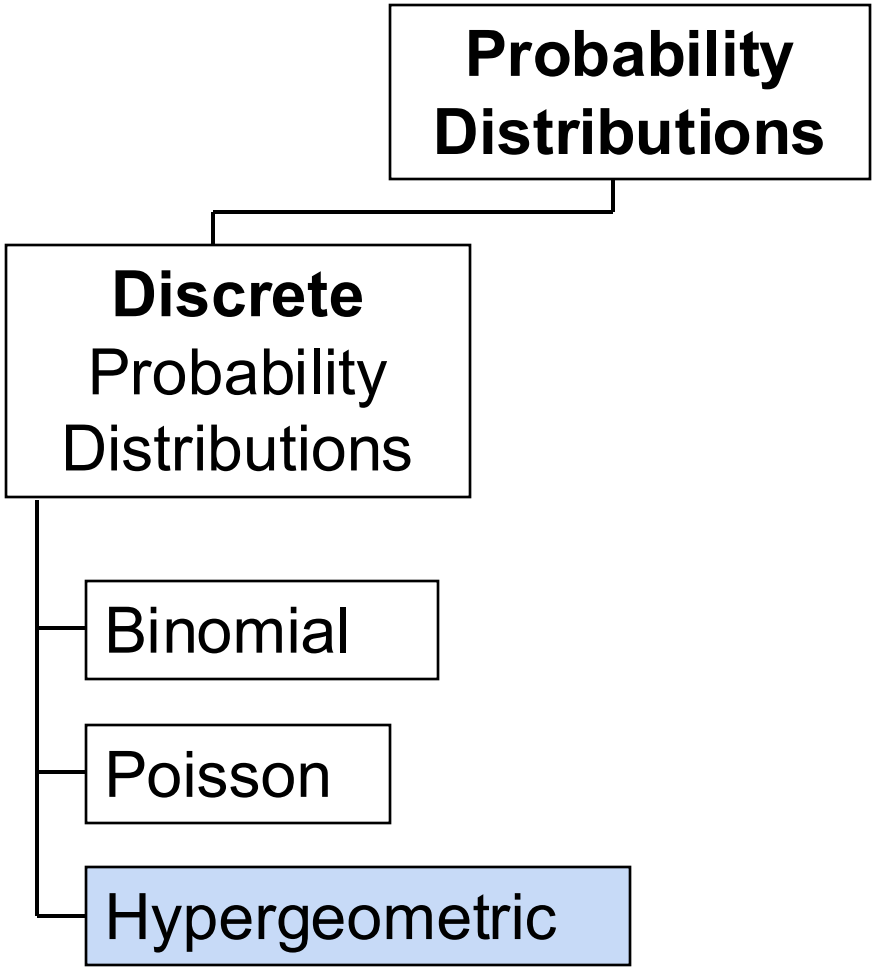


Using the Poisson Distribution

- Example: The number of faults per month that arise in the gearboxes of buses is known to follow a Poisson distribution with a mean of 2.5 faults per month.

a. What is the probability that no faults is found for a given month?
b. At least one fault is found?

The Hypergeometric Distribution



The Hypergeometric Distribution

- “n” trials in a sample taken from a finite population of size N
- Sample taken without replacement
- Outcomes of trials are dependent
- Concerned with finding the probability of “X” successes in the sample where there are “A” successes in the population

Hypergeometric Distribution Formula

$$P(X) = \frac{[{}_{A}C_{X}][{}_{N-A}C_{n-X}]}{{}_{N}C_{n}} = \frac{\binom{A}{X}\binom{N-A}{n-X}}{\binom{N}{n}}$$

Where

N = population size

A = number of successes in the population

$N - A$ = number of failures in the population

n = sample size

X = number of successes in the sample

$n - X$ = number of failures in the sample

Properties of the Hypergeometric Distribution

- The mean of the hypergeometric distribution is

$$\mu = E(x) = \frac{nA}{N}$$

- The standard deviation is

$$\sigma = \sqrt{\frac{nA(N-A)}{N^2}} \cdot \sqrt{\frac{N-n}{N-1}}$$

Where $\sqrt{\frac{N-n}{N-1}}$ is called the "Finite Population Correction Factor" from sampling without replacement from a finite population

Using the Hypergeometric Distribution

- Example: 3 different computers are checked from 10 in the department. 4 of the 10 computers have illegal software loaded. What is the probability that 2 of the 3 selected computers have illegal software loaded?

N = 10	n = 3
A = 4	X = 2

Chapter Summary

- Addressed the probability of a discrete random variable
- Defined covariance and discussed its application in finance
- Discussed the Binomial distribution
- Discussed the Poisson distribution
- Discussed the Hypergeometric distribution